

Search Engine Ranking Game Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Technology • 25-30 min

Category	Knowledge and Technology	Time	25-30 min
Difficulty	Medium	ID	search-engine-ranking-game

Big idea: Visibility is not the same as reliability.

One-Minute Teacher Brief

Students investigate how ranking affects what becomes visible as knowledge.

Student Prompt

Rank the six search results from most reliable to least reliable. Then identify which ranking signal influenced you most: expertise, popularity, freshness, sponsorship, relevance, or emotional appeal.

Concrete Stimulus Packet

Result 1: Ad · PureGreen Cleanse Co.	"7-Day Celery Juice Liver Detox — feel lighter fast."	Sells detox juice packs. No author listed. Ranking signal: Sponsorship.
Result 2: WellnessNow Blog	"I drank celery juice every morning. Here's what changed."	Personal story, dramatic before/after claims, no medical sources. Ranking signal: Emotional appeal.
Result 3: National Liver Foundation — Patient Guide	"Your liver already removes toxins from your blood."	Explains liver function; reviewed by medical advisers. Ranking signal: Expertise.
Result 4: HealthLine Daily	"Celery juice: benefits, myths, and what evidence says."	Notes hydration/nutrients but says detox claims are unsupported. Ranking signal: Relevance.
Result 5: TikTok Clip: @CleanBodyGuru	"Doctors don't want you to know this detox trick."	2.1M views, no evidence provided. Ranking signal: Popularity.
Result 6: Research Summary: Journal of Nutrition Education	"Vegetable intake and health outcomes."	Peer-reviewed; does not test celery juice detox claims directly. Ranking signal: Academic credibility, indirect relevance.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	Which result would you click first, and why?	Look for whether students choose visibility, credibility, relevance, or emotion.
2	Which result is most reliable for the query?	Likely result 3 or 4; result 6 is credible but indirect.

#	Question for students	Teacher key / cue
3	Which result might a popularity-based ranking put first?	Result 5: TikTok Clip @CleanBodyGuru.
4	Which result might a sponsorship-based ranking put first?	Result 1: PureGreen Cleanse Co. advertisement.
5	Which result best shows expertise?	Result 3: National Liver Foundation patient guide.
6	What does this show about visibility and reliability?	A ranking system can make a weak result visible before a strong one.

Student Task

Query: Does drinking celery juice detox your liver? Rank all six results from most reliable to least reliable, then explain which signal shaped your ranking.

Teacher Key / Reveal

A search engine could rank result 5 highest by popularity, result 1 highest by sponsorship, result 4 highest by relevance, and result 3 highest by expertise. Visibility is not the same as reliability.

- Most medically reliable: Result 3, then Result 4.
- Most popular-looking: Result 5.
- Most commercially promoted: Result 1.
- Most academically credible but indirect: Result 6.
- Least reliable for knowledge: Result 5 or Result 2, depending on criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Does visibility affect what people accept as true?

Setup Checklist

- Prepare a starter stimulus using: Mock search results ranked by popularity, expertise, freshness, sponsorship, or relevance.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- Mock search results ranked by popularity, expertise, freshness, sponsorship, or relevance.
- A query about a health claim with credible, vague, commercial, and opinion sources.
- Two ranking rules that produce different 'top answers' for the same searcher.

Resources To Use

- Resource pack: <resources/search-engine-ranking-game/index.html>
- Card deck CSV: <resources/search-engine-ranking-game/cards.csv>
- Visual prompt: <resources/search-engine-ranking-game/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.

- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Present a question such as "Is homework effective?"
2. Students rank which results they would click first.
3. Reveal hidden information: sponsored result, old source, expert review, blog post, advocacy group and research article.
4. Students redesign the search page to make reliability easier to judge.

Debrief Moves

- Knowledge question: Does visibility affect what people accept as true?
- Is the first answer usually the best answer?
- How should search engines balance relevance, profit and public knowledge?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...